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Assignment No. 1 (Part-I)

Aim: Implement Union, Intersection, Complement and Difference operations on fuzzy sets. Also create fuzzy relations by Cartesian product of any two fuzzy sets and perform max min composition on any two fuzzy relations.

Theory:

Logic is a tool for reasoning propositions that can be manipulated with mathematical precepts. Fuzzy Logic: The word fuzzy means uncertainty. Any particular event which do not result any of the exact value (i.e. true or false) is fuzzy. A proposition is a declarative or linguistic statement within a universe of discourse. For example, Elizabeth is tall. In classical logic a proposition is either true or false. That means the proposition 'Elizabeth is tall' can be either true or false.

shows a real-world situation where the glass is more than half full of water.

Fuzzy logic is a transition from absolute truth to partial truth. That is, from a variable x (True or False) to a linguistic variable 'Almost full', 'Very close to empty', etc. From this perspective, fuzzy logic can be seen as a reasoning formalism of humans where all truths are partial or approximate and any falseness is represented by partial truth

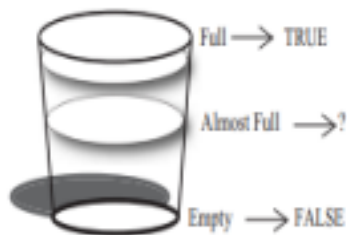
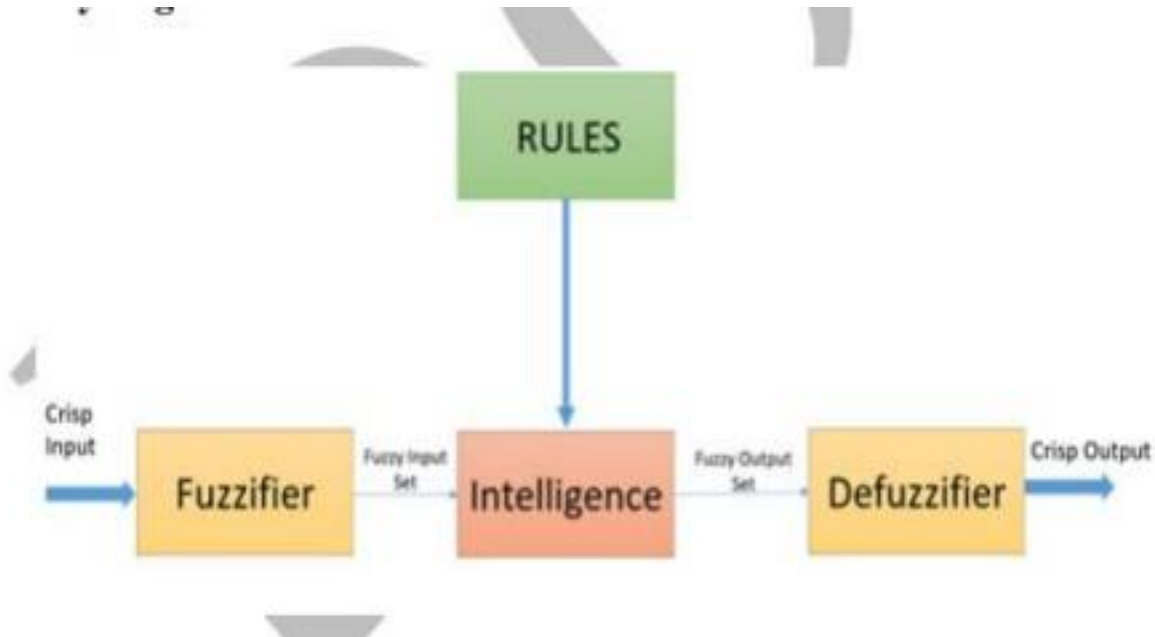


Figure 2.1 A real-world situation



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Rule Base: It contains all the rules and the if-then conditions offered by the experts to control the decision-making system.

Inference Engine: It helps you to determine the degree of match between fuzzy input and the rules. Based on the % match, it determines which rules need implement according to the given input field. After this, the applied rules are combined to develop the control actions.

Fuzzification step helps to convert inputs. It allows you to convert, crisp numbers into fuzzy sets. Crisp inputs measured by sensors and passed into the control system for further processing. Like Room temperature, pressure, etc.

Defuzzification: At last the Defuzzification process is performed to convert the fuzzy sets into a crisp value. There are many types of techniques available, so you need to select it which is best suited when it is used with an expert system.

For example, For example, a fuzzy set $A = \{x_1, x_2, x_3, x_4\}$ in X is characterized by the membership function $\mu_A(x)$ which maps each point x in X to real values 0.5, 1, 0.75 and 0.5. $\mu_A(x)$ represents the degree of membership of x in A and the mapping is only limited by $\mu_A(x) \in [0, 1]$. In classical set theory, the membership function can take only two values: 0 and 1, i.e., either $\mu_A(x)=1$ or $\mu_A(x) = 0$. In set-theoretic notation this is written as $\mu_A(x) \in \{0, 1\}$. A fuzzy set is an extension of a



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classical set. If X is the universe of discourse and its elements are denoted by x , then a fuzzy set A in X is defined as a set of ordered pairs

$$A = \{x, \mu_A(x) \mid x \in X\}$$

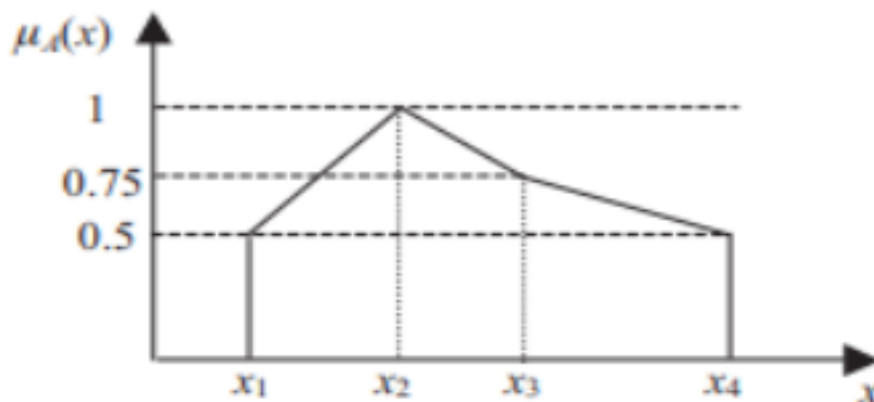


Figure 2.2 Fuzzy set

This mapping can be depicted pictorially, as shown in Figure 2.2. In Figure 2.2, x_1 , x_2 , x_3 and x_4 have membership grades of 0.5, 1, 0.75 and 0.5, respectively, written as $\mu_A(x_1) = 0.5$, $\mu_A(x_2) = 1$, $\mu_A(x_3) = 0.75$ and $\mu_A(x_4) = 0.5$. A notational

convention of fuzzy sets for a discrete and finite universe of discourse X in practice is written as $A = \{\mu_A(x_1)/x_1 + \mu_A(x_2)/x_2 + \dots + \mu_A(x_n)/x_n\} = \sum_{i=1}^n \mu_A(x_i)/x_i$

Fuzzy Rules:

R: If x is A , then y is B .

where A and B are linguistic values defined by fuzzy sets on universes of discourse X and Y , respectively.

The rule is also called a “fuzzy implication” or fuzzy conditional statement. The part “ x is A ” is



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called the “antecedent”, while “y is B” is called the “consequence” or “conclusion”. It typically expresses an inference such that if we know a fact or a hypothesis (antecedent,), then we can infer, or derive, another fact called a conclusion (consequent)..

Before we employ fuzzy if-then rules to model and analyse a system, first we have to formalize what is meant by the expression: R: If x is A then y is B which is sometimes abbreviated as

$$R: A \rightarrow B$$

Conclusion: -

- Questions:

Q1. What FUZZY?explain with Ex.?

Q2. Fuzzy Application ?

Q3. Explain Fuzzy Architecture?

Q4. Differentiate Classical Vs Fuzzy Rule?